

AI-Powered Customer Support Chatbot – project overview

Business case :

- The Business needs to handle the minute ,less important enquiries of the user that leads to hiring additional support staff and increase the cost ,static FAQ pages also doesn't reduce the traffic.
- So to improve this we can utilize AI we can build a AI Chatbot that helps the user to ask about most commonly asked FAQ,order status,cancellation and refund.It can help the customer care reduce workload and time around 5% in a month.

Crisp-dm :

Crisp-dm stages	What happens	What BA owns
Business requirements	Define why we are building a AI Chatbot .	Collects the business requirements such as type of user and,business case scenario and success metric of the business outcome like need to reduce 5% in workload for customer care in a month .
Data requirements	Collect all types of internal data and external data of customers.	Have customer data, order data,customer feedback data.
Data preparation	The collected raw data is cleaned, prepared,checked for anomalies and gaps in the data for the analysis of the find the doubts and enquiries the user have to help them.	Give a logic and what business need more of like a BRD to the data developers to know what are the data they needs to prepare.
Modeling	Then the cleaned customer data is in ready to use then we do predictive,nonpredictive ,statistical,descriptive analysis to model the data.	Give a details about the constraints that the data they use for training the chat bot should not contain any personal information of customer only about the order id ,order status and most commonly asked FAQ also developers have to know financial constraints and others. And also needs to

		know the model they are using and translate it to the stakeholders.
Evaluating	Then we test and evaluate the chatbot with various categories and check whether they give a quality output for all type of user and all type of enquiries.	Checks the developed model whether it meets the business requirements and business use case also checks with the stakeholder and the success criteria of the model like will chatbot able to give answer correctly.
Deployment	Then after modelling and evaluating and test the chatbot and improve the needs and deploy it.	Prepare all the necessary documents that helps with the customer to learn how their model works and also get them informed that they are communicating with AI and if need they can communicate with human customer care person and also stay connected with the stakeholder to the necessary change and improvement in the model.

Success criteria :

- The chatbot should have tested and answer all the approved question around 95%.
- Customer satisfaction around 80%.
- Human escalation question should be less around 80%.
- Note: 90% accuracy and 80% precision are the minimum model requirements for launch (UAT gate). The 95% success criterion is the target to be reached after post-launch tuning.

Edge case :

- When questions are not in understandable format.
- The question that are out of approved dataset.
- Questions that have one or more answers.
- Involvement of the human customer support.

Constraints :

- Chatbot should not generate answer on its own.
- The bot should not ask anything that related to data privacy of a user.
- The chatbot should not retain any credit card/debit card/upi/net banking of a user.

Risks and Ethics :

Risk	The chatbot ask for account password for user order enquiry.
Impact	This can cause privacy issue for the user.
Mitigation	collect and process required data need to know about the order eg.(order id,order status).

Risk	The chatbot training data is less for FAQ of customer common doubts.
Impact	It increase the workload for customer care and time consuming for business.
Mitigation	collect and process more data about common FAQ.

Risk	The user doesn't know that they are communicating with AI.
Impact	The user may misunderstood and rely on its responses.
Mitigation	Inform the user they are communicating with AI and if need provide human escalation.

Risk	The chatbot training data may contain mostly formal English customer queries and fewer examples of informal wording or different supported languages.
Impact	The chatbot may understand some customers better than others and provide poorer support to certain users.
Mitigation	Use representative customer-query data and test chatbot performance across different supported languages and communication styles.